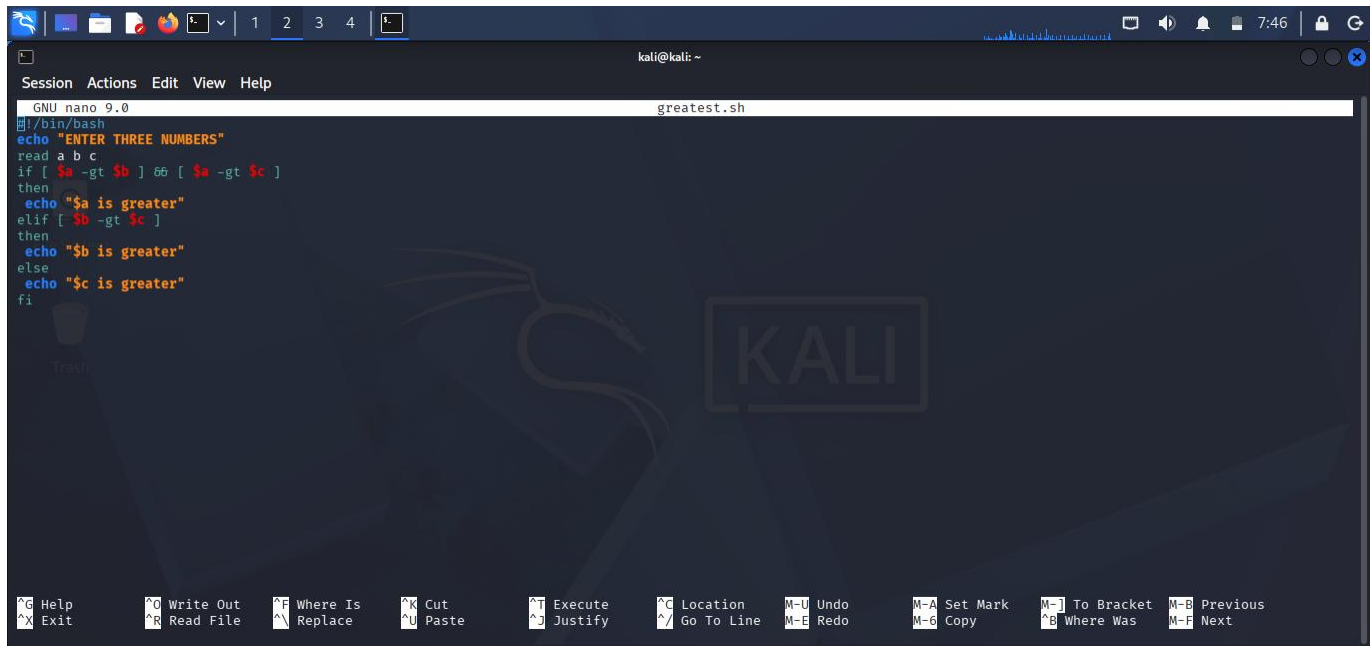


SHELL PROGRAMMING :GREATEST AMONG THREE NUMBERS

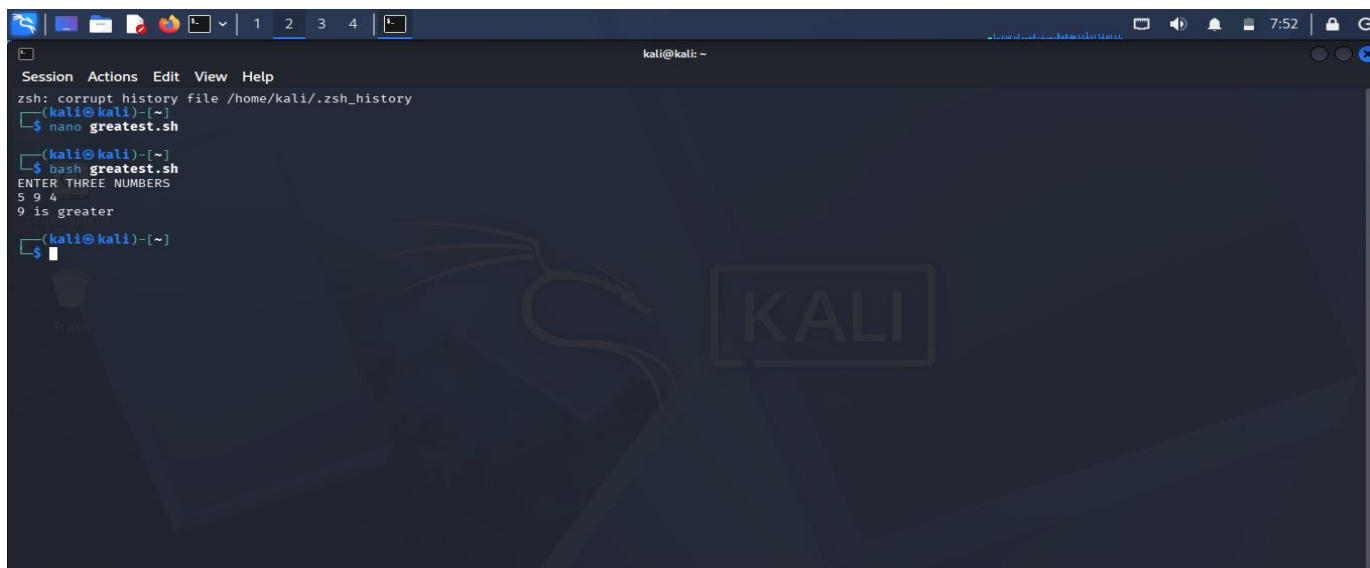


The screenshot shows a terminal window with the nano text editor open. The editor is editing a file named 'greatest.sh'. The script content is as follows:

```
#!/bin/bash
echo "ENTER THREE NUMBERS"
read a b c
if [ $a -gt $b ] && [ $a -gt $c ]
then
    echo "$a is greater"
elif [ $b -gt $c ]
then
    echo "$b is greater"
else
    echo "$c is greater"
fi
```

The terminal window has a dark theme with a Kali Linux logo in the background. The nano editor's menu bar includes Session, Actions, Edit, View, and Help. The status bar at the bottom shows various keyboard shortcuts for editing and navigation.

OUTPUT :

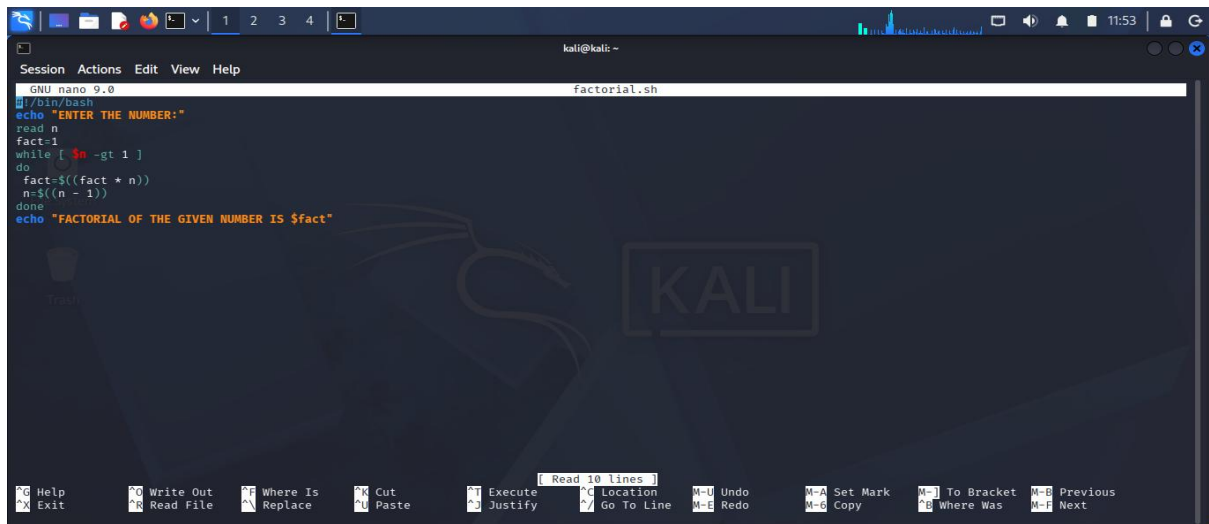


The screenshot shows the terminal window after running the script. The output is as follows:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ nano greatest.sh
(kali@kali)-[~]
$ bash greatest.sh
ENTER THREE NUMBERS
5 9 4
9 is greater
(kali@kali)-[~]
$
```

The terminal window shows the script being executed. The user enters three numbers (5, 9, 4) and the script outputs '9 is greater'. The terminal window has a dark theme with a Kali Linux logo in the background.

SHELL PROGRAMMING :FACTORIAL OF A GIVEN NUMBER

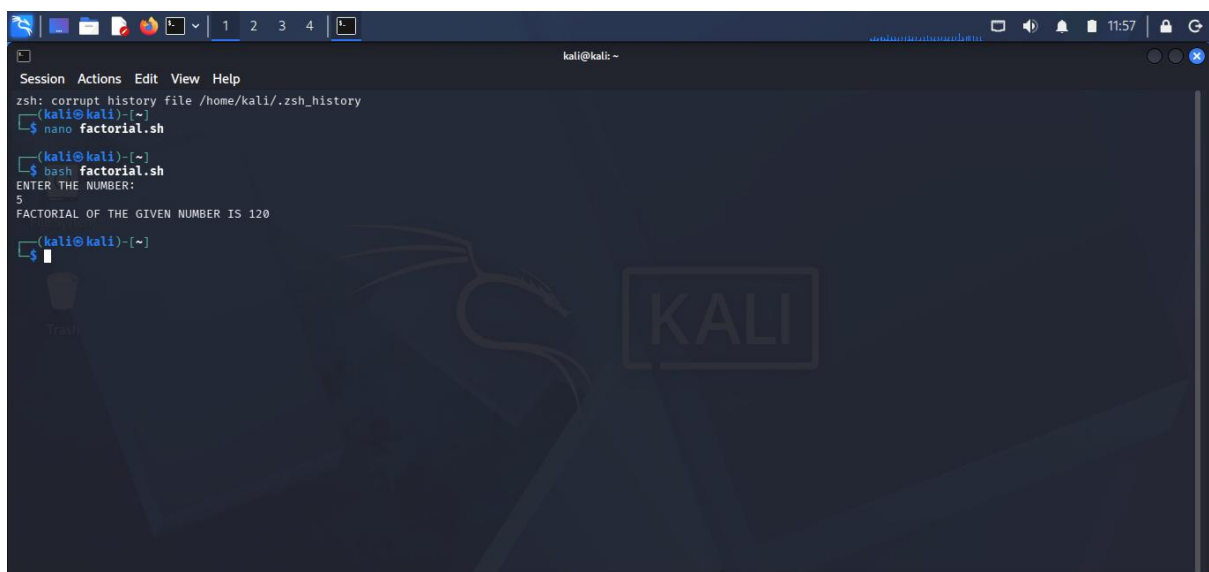


The screenshot shows a terminal window with the nano text editor open. The editor is editing a file named 'factorial.sh'. The script contains the following code:

```
#!/bin/bash
echo "ENTER THE NUMBER:"
read n
fact=1
while [ $n -gt 1 ]
do
    fact=$((fact * n))
    n=$((n - 1))
done
echo "FACTORIAL OF THE GIVEN NUMBER IS $fact"
```

The terminal window has a title bar that says 'kali@kali: ~'. The nano editor's status bar at the bottom shows various keyboard shortcuts like 'Help', 'Exit', 'Write Out', 'Read File', 'Where Is', 'Replace', 'Cut', 'Paste', 'Execute', 'Justify', 'Location', 'Go To Line', 'Undo', 'Redo', 'Set Mark', 'Copy', 'To Bracket', 'Where Was', 'Previous', and 'Next'.

OUTPUT :

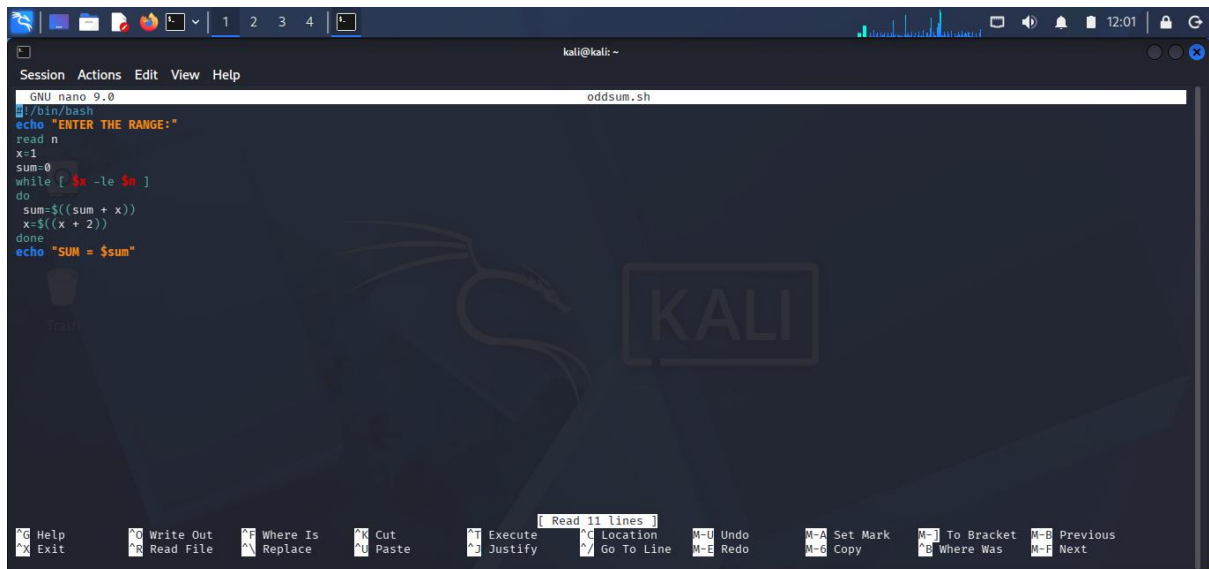


The screenshot shows a terminal window with the following output:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ nano factorial.sh
(kali@kali)~$ bash factorial.sh
ENTER THE NUMBER:
5
FACTORIAL OF THE GIVEN NUMBER IS 120
(kali@kali)~$
```

The terminal window has a title bar that says 'kali@kali: ~'. The background of the terminal window features a Kali Linux logo and the word 'KALI'.

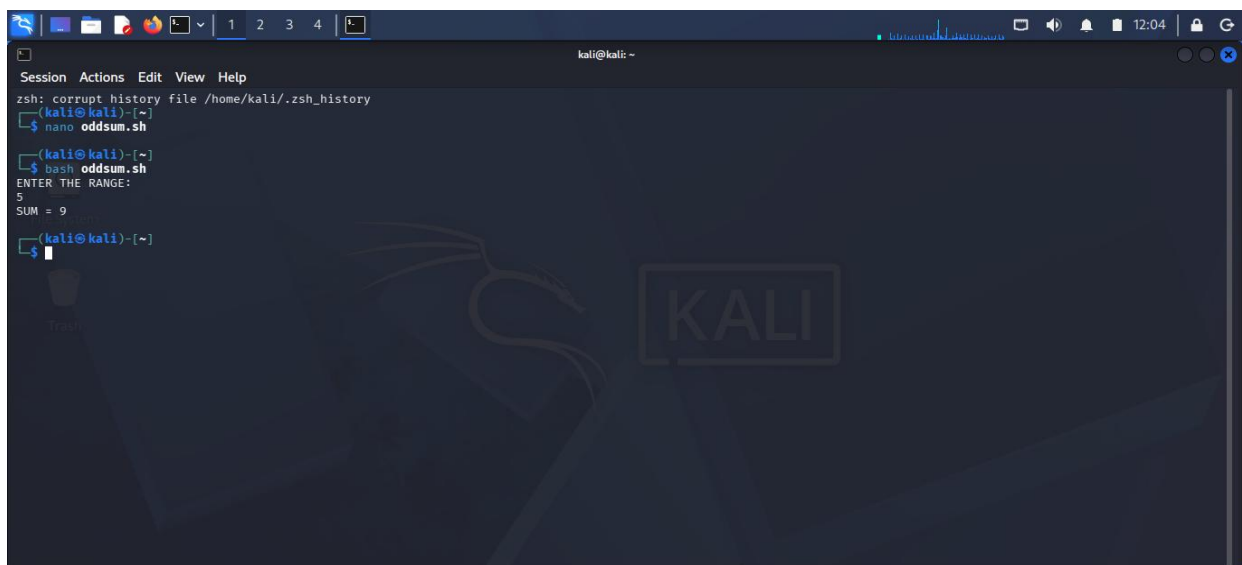
SHELL PROGRAMMING :SUM OF ODD NUMBERS



```
GNU nano 9.0 oddsum.sh
#!/bin/bash
echo "ENTER THE RANGE:"
read n
sum=0
x=1
while [ $x -le $n ]
do
    sum=$((sum + x))
    x=$((x + 2))
done
echo "SUM = $sum"
```

The screenshot shows a terminal window with the nano text editor open. The editor is editing a file named 'oddsum.sh'. The script is a bash script that prompts the user to enter a range, reads the input, and then calculates the sum of odd numbers from 1 to that range using a while loop. The background of the terminal has a Kali Linux logo watermark.

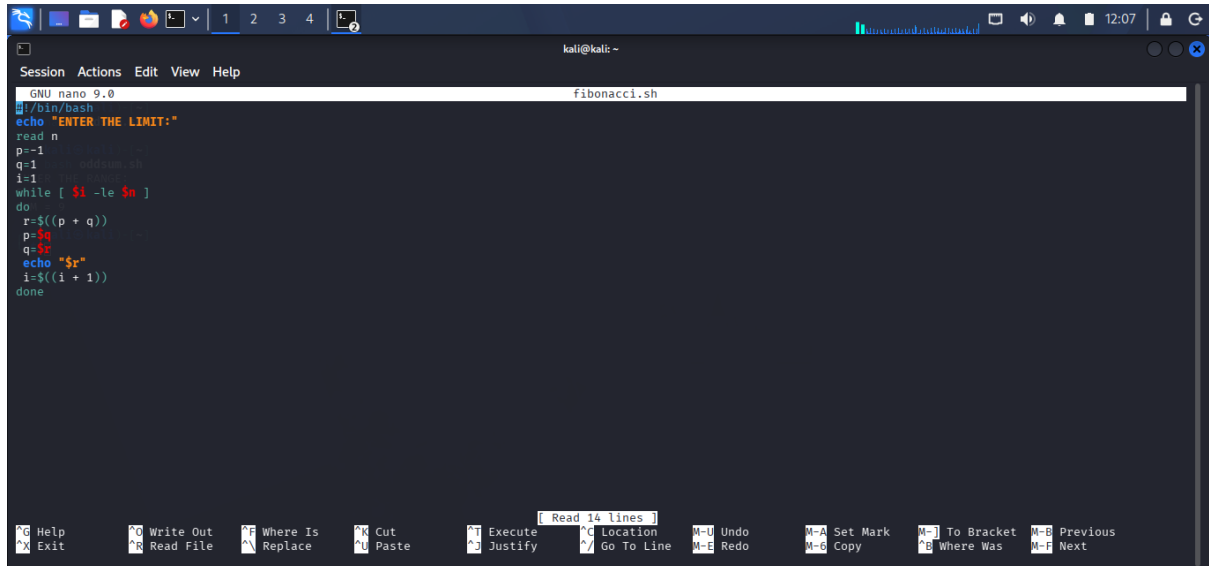
OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
└─$ nano oddsum.sh
(kali@kali)-[~]
└─$ bash oddsum.sh
ENTER THE RANGE:
5
SUM = 9
(kali@kali)-[~]
└─$
```

The screenshot shows the terminal output of the script. The user has run 'bash oddsum.sh', which prompted them to enter a range. They entered '5', and the script outputted 'SUM = 9'. The terminal window also shows the history of commands and the current directory is '~' (home).

SHELL PROGRAMMING :FIBONACCI NUMBER

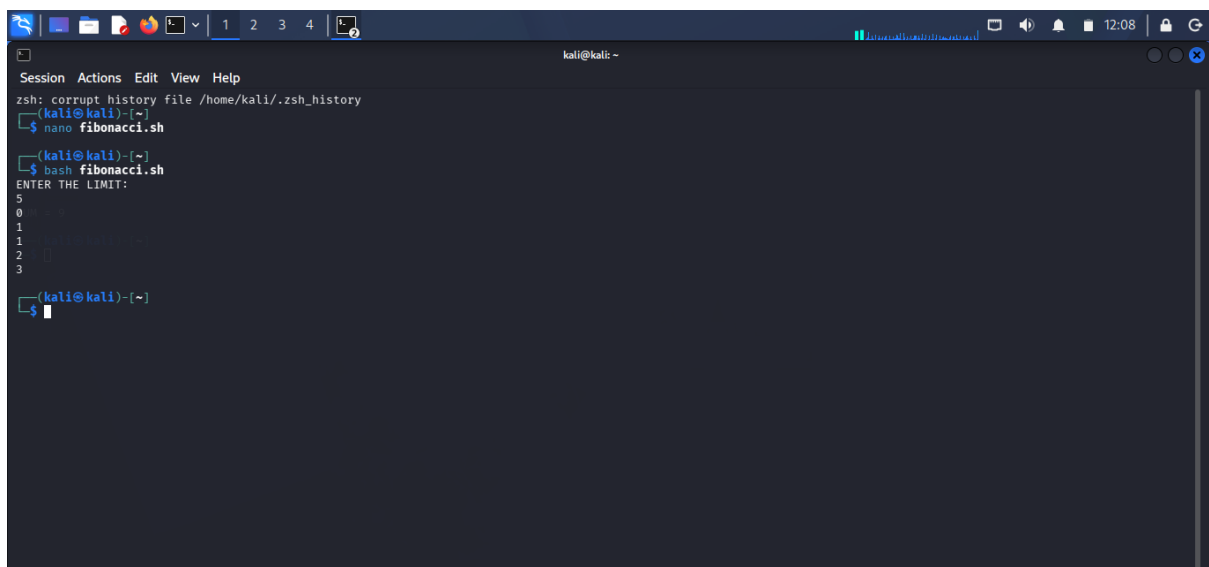


A screenshot of a terminal window on a Kali Linux system. The terminal shows the nano 9.0 text editor editing a file named 'fibonacci.sh'. The script content is as follows:

```
#!/bin/bash
echo "ENTER THE LIMIT:"
read n
p=1
q=1
i=1
while [ $i -le $n ]
do
r=$((p + q))
p=$q
q=$r
echo "$r"
i=$((i + 1))
done
```

The terminal window includes a menu bar with options: Session, Actions, Edit, View, Help. At the bottom, there is a status bar with various keyboard shortcuts like Help, Exit, Write Out, Read File, Where Is, Replace, Cut, Paste, Execute, Justify, Location, Go To Line, Undo, Redo, Set Mark, Copy, To Bracket, Where Was, Previous, Next.

OUTPUT :

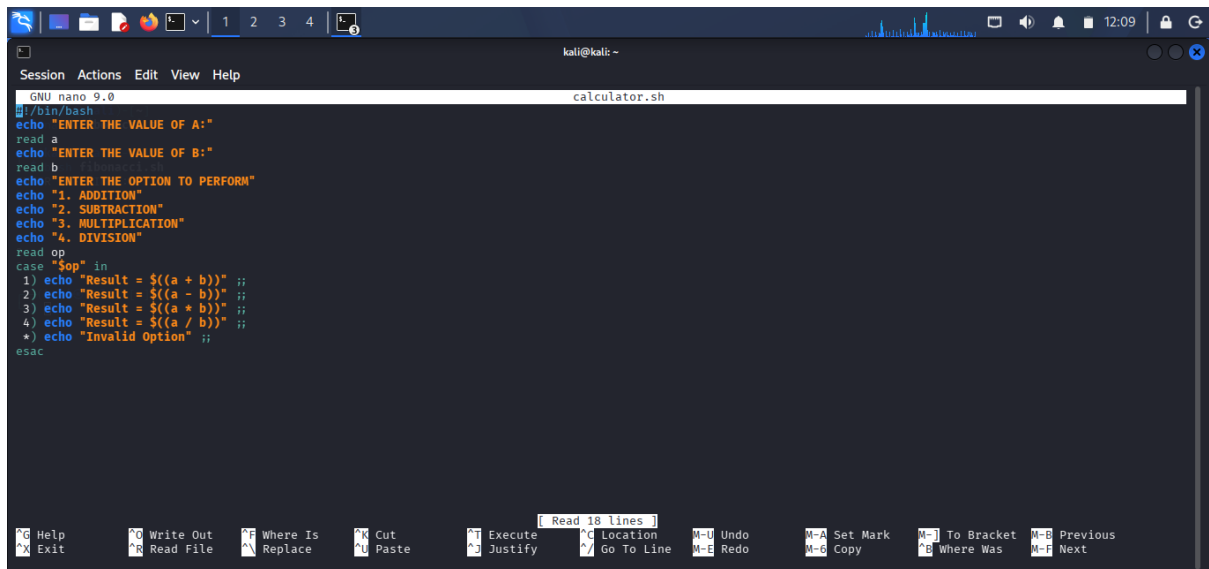


A screenshot of a terminal window showing the execution of the 'fibonacci.sh' script. The terminal output is as follows:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ nano fibonacci.sh
(kali@kali)~$ bash fibonacci.sh
ENTER THE LIMIT:
5
0
1
1
2
3
```

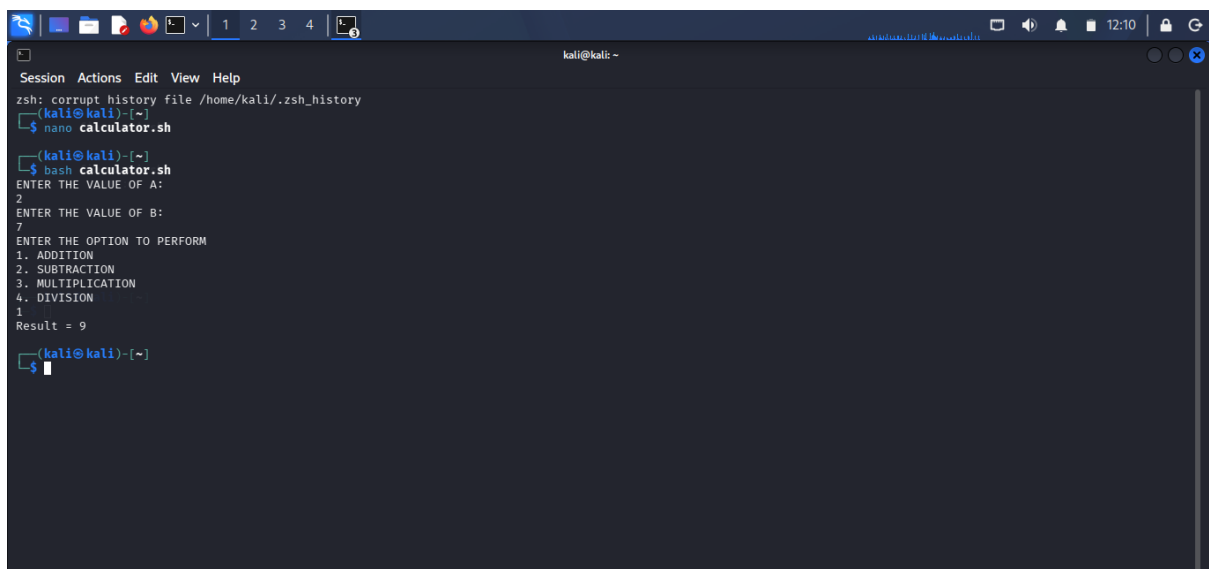
The terminal window shows the user entering the limit '5' and the script outputting the Fibonacci sequence: 0, 1, 1, 2, 3. The terminal window includes a menu bar with options: Session, Actions, Edit, View, Help. At the bottom, there is a status bar with various keyboard shortcuts like Help, Exit, Write Out, Read File, Where Is, Replace, Cut, Paste, Execute, Justify, Location, Go To Line, Undo, Redo, Set Mark, Copy, To Bracket, Where Was, Previous, Next.

SHELL PROGRAMMING :ARITHMETIC CALCULATOR



```
GNU nano 9.0 calculator.sh
#!/bin/bash
echo "ENTER THE VALUE OF A:"
read a
echo "ENTER THE VALUE OF B:"
read b
echo "ENTER THE OPTION TO PERFORM"
echo "1. ADDITION"
echo "2. SUBTRACTION"
echo "3. MULTIPLICATION"
echo "4. DIVISION"
read op
case "$op" in
1) echo "Result =  $$(a + b)$ " ;;
2) echo "Result =  $$(a - b)$ " ;;
3) echo "Result =  $$(a * b)$ " ;;
4) echo "Result =  $$(a / b)$ " ;;
*) echo "Invalid Option" ;;
esac
```

OUTPUT :



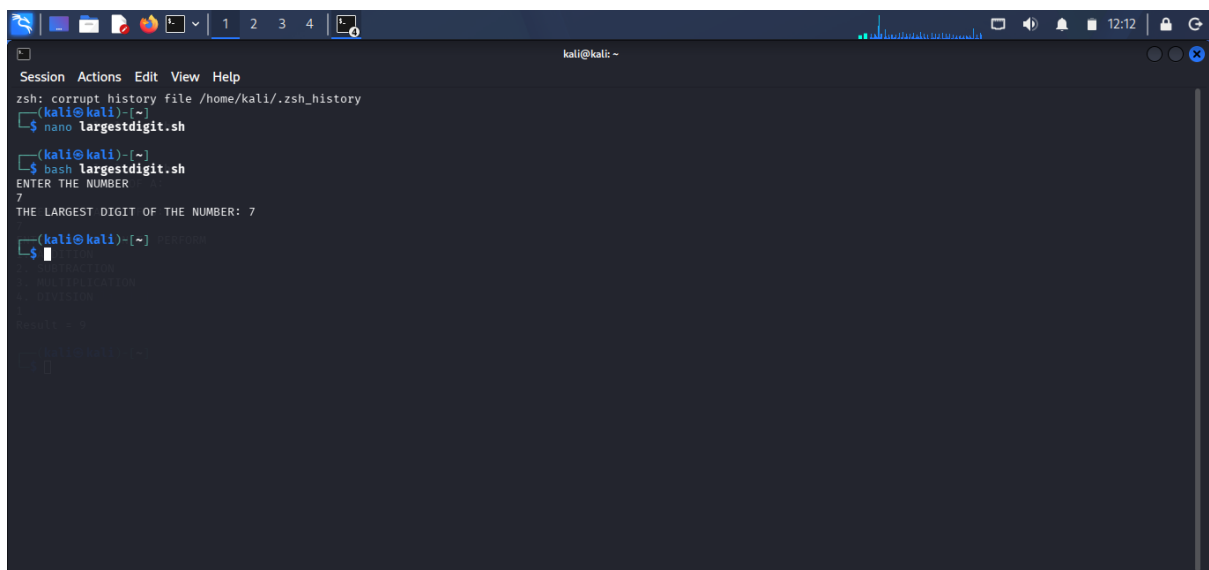
```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ nano calculator.sh
(kali@kali)~$ bash calculator.sh
ENTER THE VALUE OF A:
2
ENTER THE VALUE OF B:
7
ENTER THE OPTION TO PERFORM
1. ADDITION
2. SUBTRACTION
3. MULTIPLICATION
4. DIVISION
1
Result = 9
(kali@kali)~$
```

SHELL PROGRAMMING :LARGEST DIGIT OF A NUMBER



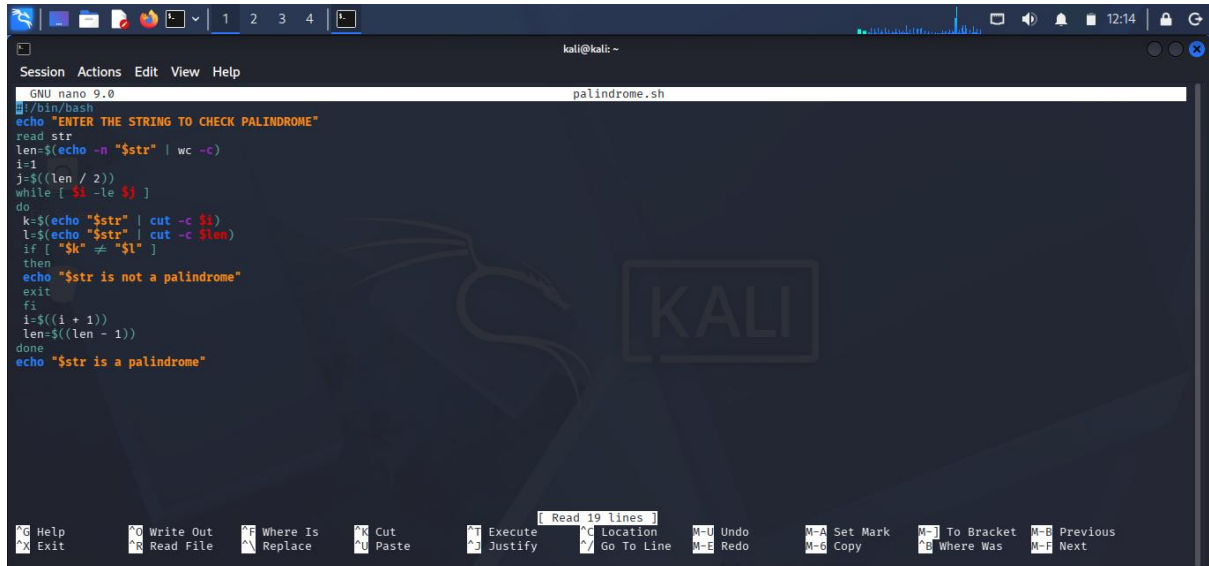
```
GNU nano 9.0 largestdigit.sh
#!/bin/bash
echo "ENTER THE NUMBER"
read a
max=0
while [ $a -gt 0 ]
do
    r=$((a % 10))
    if [ $r -gt $max ]
    then
        max=$r
    fi
    a=$((a / 10))
done
echo "THE LARGEST DIGIT OF THE NUMBER: $max"
```

OUTPUT :



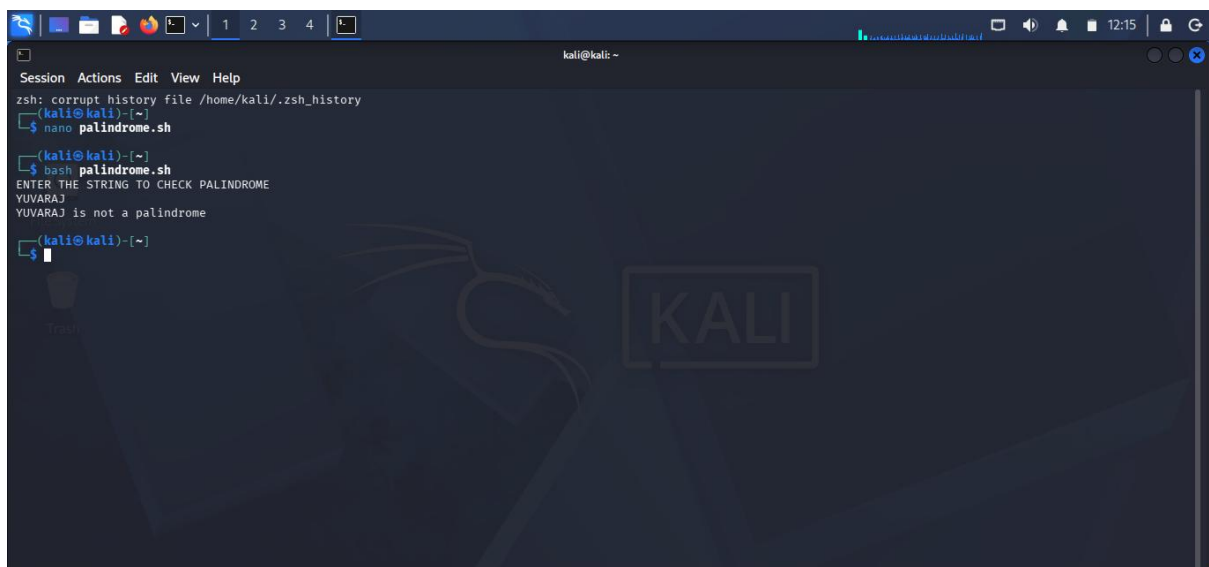
```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ nano largestdigit.sh
(kali@kali)~$ bash largestdigit.sh
ENTER THE NUMBER
7
THE LARGEST DIGIT OF THE NUMBER: 7
(kali@kali)~$
```

SHELL PROGRAMMING :PALINDROME OR NOT



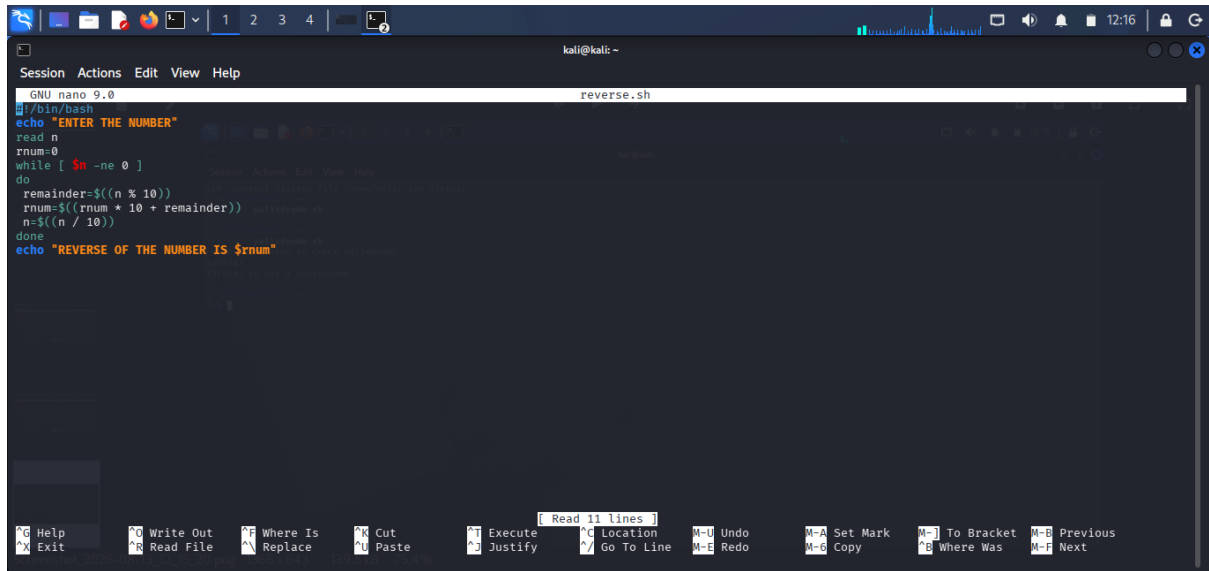
```
GNU nano 9.0
#!/bin/bash
echo "ENTER THE STRING TO CHECK PALINDROME"
read str
len=$(echo -n "$str" | wc -c)
i=1
j=$((len / 2))
while [ $i -le $j ]
do
k=$(echo "$str" | cut -c $i)
l=$(echo "$str" | cut -c $len)
if [ "$k" != "$l" ]
then
echo "$str is not a palindrome"
exit
fi
i=$((i + 1))
len=$((len - 1))
done
echo "$str is a palindrome"
```

OUTPUT :



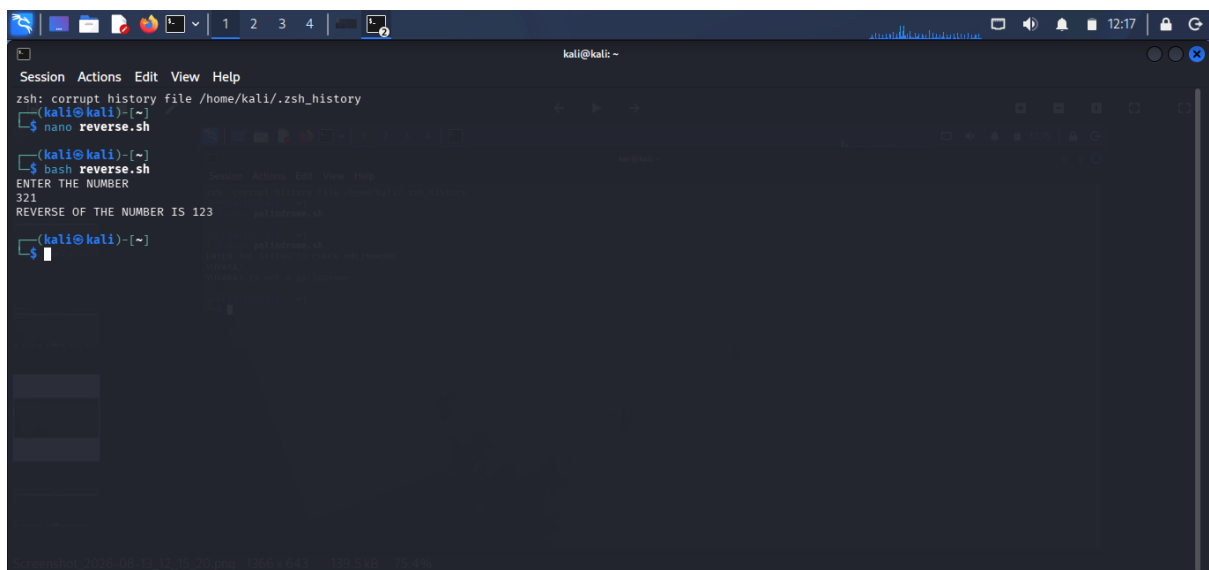
```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ nano palindrome.sh
(kali@kali)~$ bash palindrome.sh
ENTER THE STRING TO CHECK PALINDROME
YUVARAJ
YUVARAJ is not a palindrome
(kali@kali)~$
```

SHELL PROGRAMMING :REVERSE OF A GIVEN NUMBER



```
GNU nano 9.0 reverse.sh
#!/bin/bash
echo "ENTER THE NUMBER"
read n
rnum=0
while [ $n -ne 0 ]
do
    remainder=$((n % 10))
    rnum=$((rnum * 10 + remainder))
    n=$((n / 10))
done
echo "REVERSE OF THE NUMBER IS $rnum"
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
└─$ nano reverse.sh
(kali@kali)-[~]
└─$ bash reverse.sh
ENTER THE NUMBER
321
REVERSE OF THE NUMBER IS 123
(kali@kali)-[~]
└─$
```